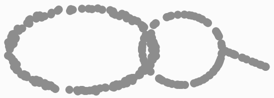
Lithium — label the surface of anatomy, train in the loop, read the frame off the labels

1 Surface point clouds



32,000 points per bone
instead of ~500k voxels
247 VerSe vertebrae · 12 subjects

2 Paint labels on the surface

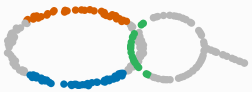
- superior endplate
- inferior endplate
- pedicle

pick 27 ms at 1 M points

3 Train PT-v3 in the app

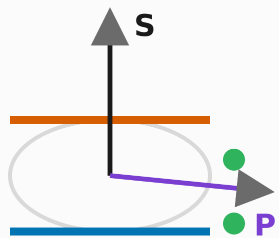
catalog = dataset
↓
export PT-v3 scenes
↓
train (sidecar process)
↓
model registry

4 Predict, correct, repeat



INFER: 2.4 s per bone
on one RTX 4090
correct, retrain, run again

5 Anatomy from labels



endplate normals → S
pedicle pair → P, L = S × P

Held-out results

38 bones, subjects unseen in training

endplate precision
0.94 / 0.92
pedicle precision
0.63 ← the open gap
axis error (median)
0.39°
centre error (median)
0.6 mm

Open source (MIT) · Python · one GPU workstation · no cloud, no accounts · github.com/saffffari/Lithium